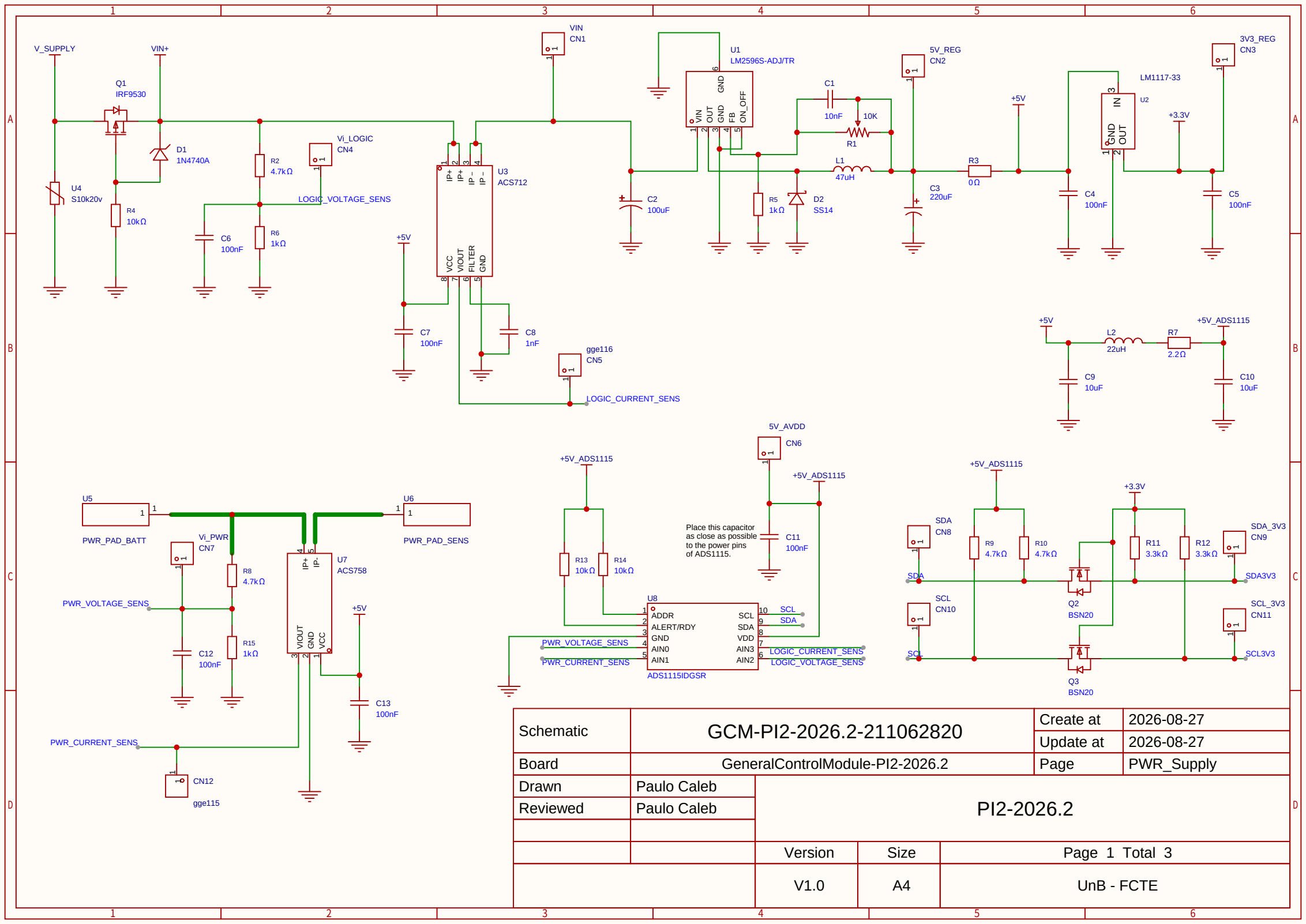
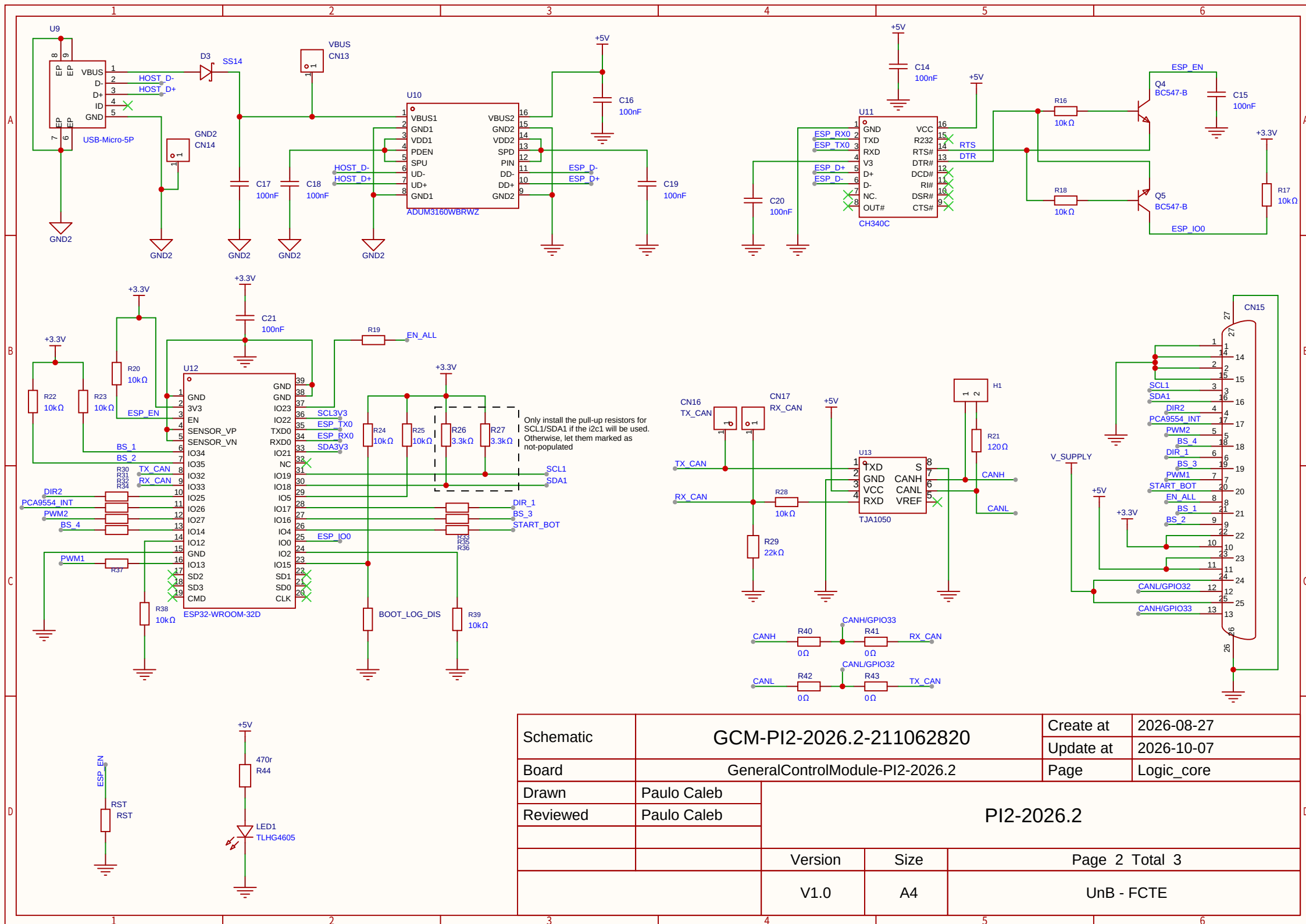




Schematic	GCM-PI2-2026.2-211062820			Create at	2026-08-27
				Update at	2026-08-27
Board	GeneralControlModule-PI2-2026.2			Page	PWR_Supply
Drawn	Paulo Caleb	PI2-2026.2			
Reviewed	Paulo Caleb				
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Design Notes:

Even though the ESP32-WROOM-32D is marked as NRND (Not Recommended for New Designs), it was selected for the GCM board due to its large availability in the domestic market and low cost in Brazil;

The same criteria were used to select the LM2596 buck converter, TJA1050 CANbus transceiver, ACS712/ACS758 hall-effect current sensors, and CH340 USB-TTL bridge. Better options aren't easily available, so the best available choice was selected;

Other components, such as the ADuM3160 and BSN20, were selected for their functionality in the application, making it inevitable to import these components from the international market;

In the simulation to the side, it's possible to see the behavior of the pi filter designed to filter the supply rail of the ADS1115 analog-to-digital converter;

The simulation was performed in LTspice software, and the simulated components match the real specs of the components sold in the market; (see: datasheets/capacitor_filtro_pi)

Based on the simulation, it is possible to see that the behavior of the filter is sufficiently efficient to attenuate the high frequency ripple generated by the LM2596 at 150kHz fixed switching frequency (-56dB). Thus it is expected that this ripple will not affect the analog section of the circuit;

The decoupling capacitors are meant to be placed as close as possible to the ICs.

The four mounting holes are intended to contact the robot's chassis for grounding redundancy. Leaving these GND points disconnected is not a issue, as the main connector includes redundant GND lines, unless otherwise specified by the application.

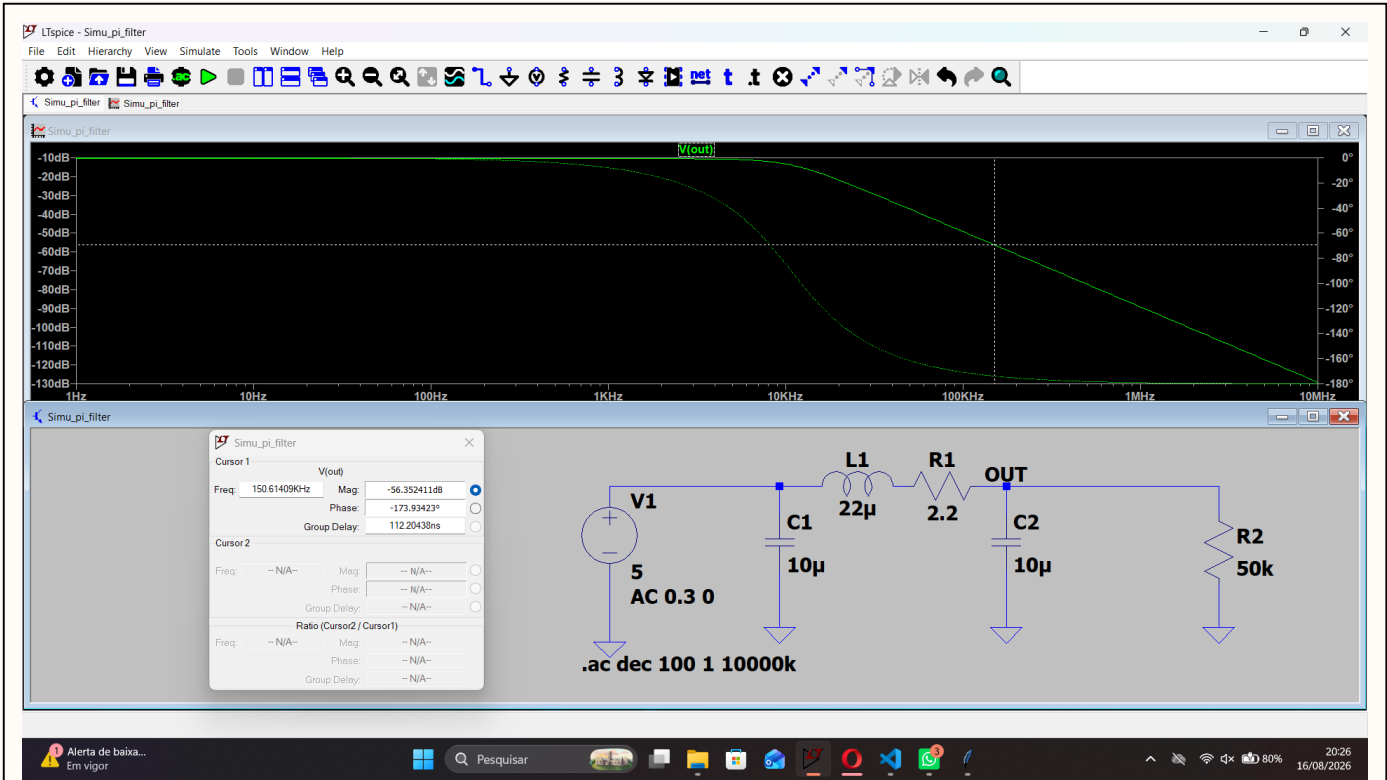
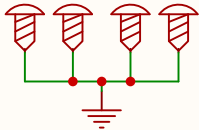


Image 1 - Screen capture from the simulation of the pi-filter designed to filter the supply rail of the analog-to-digital converter used in the power telemetry system of the board

THEORICAL CALCULATIONS:

Cutoff frequency of pi filter:
 $f_c = 1/2 \pi \sqrt{LC} = 1/2 \pi \sqrt{22 \mu H \times 10 \mu F} \approx 10.7 kHz$
Attenuation @ LM2596's switching frequency (150kHz):
 $40 \times \log_{10}(150 kHz / 10.7 kHz) \approx 45.8 dB \approx 195 \times$

SIMULATED BEHAVIOR:
Attenuation @ 150kHz in the LTspice software: 56dB

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